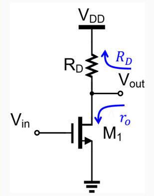
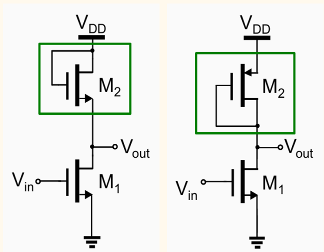
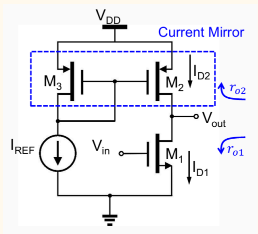
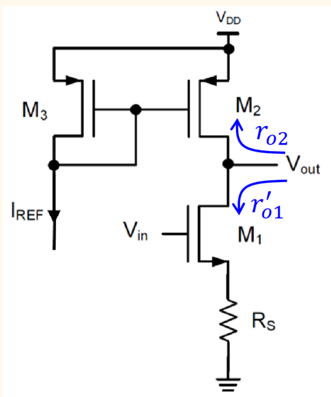
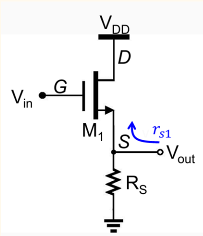
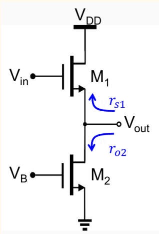
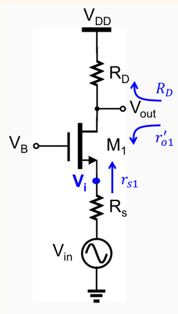
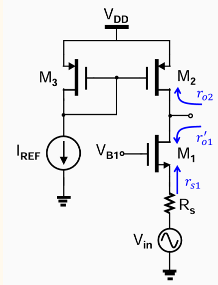
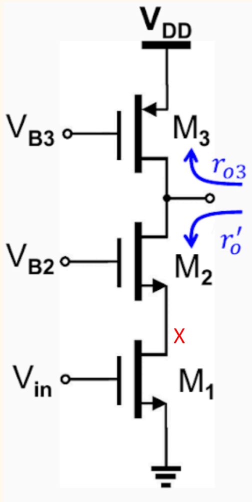
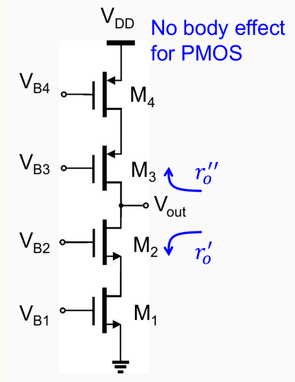
Very Important Table

Topology		Circuit	Voltage Gain, $A_V = -G_m r_o'$	Rout = r_o'	Rin = r_g or r_s	Remarks
CS	R_D		$A_v = -g_{m1}(r_{o1}\|\|R_D)$	$r_{o1}\|\|R_D$	∞	Rail to rail
	Diode		$A_v = -g_{m1}(r_{o1}\|\|r_{diode})$ $= -g_{m1}(r_{o1}\|\|\frac{1}{g_{m2}+g_{mb2}}\|\|r_{o2})$ $\approx -g_{m1}\frac{1}{g_{m2}+g_{mb2}}$ $= -\frac{g_{m1}}{g_{m2}}\frac{1}{1+\eta}$ $= -\sqrt{\frac{(W/L)_1}{(W/L)_2}}\frac{1}{1+\eta}$ **ONLY IF BOTH ARE N/P, else kn/p cannot cancel**	$r_{o1}\|\|r_{diode}$ $= r_{o1}\|\|\frac{1}{g_{m2}+g_{mb2}}\|\|r_{o2}$ $\approx \frac{1}{g_{m2}+g_{mb2}}$	∞	Top swing limited by $V_{DD} - \|V_{TH2}\|$
	CSL		$A_v = -g_{m1}(r_{o1}\|\|r_{o2})$	$r_{o1}\|\|r_{o2}$	∞	For mid point output, $I_{D1} = I_{D2}$ CSL to avoid tradeoff between gain and voltage swing
	CSL (degen)		$G_m = \frac{g_{m1}\frac{1}{R_s}}{g_{m1}+g_{mb1}+\frac{1}{R_s}+\frac{1}{r_{o1}}}$ $= \frac{r_{o1}g_{m1}}{r_{o1}R_s(g_{m1}+g_{mb1})+r_{o1}+R_s}$ (same as CD since Vout is grounded for Gm calculation) If $g_{m1}+g_{mb1} \gg \frac{1}{R_s}$ & $\frac{1}{r_{o1}}$, $G_m \approx \frac{1}{(1+\eta)R_s}$	$r_{o1}'\|\|r_{o2}$ If $g_{m1}+g_{mb1} \gg \frac{1}{R_s}$, $\approx r_{o2}$	∞	Degeneration resistors makes the Gm independent of gm, improving linearity of gain Degeneration makes voltage gain to be = (R the drain sees)/(R the source sees) = top/bottom for NMOS = bottom/top for PMOS

			$r_{o1}' = [1 + \frac{R_S}{r_{o1}} + (g_{m1} + g_{mb1})R_S]r_{o1}$ $A_v = G_m r_o'$ $= \frac{-r_{o1}r_{o2}g_{m1}}{r_{o1}R_s(g_{m1}+g_{mb1})+r_{o1}+r_{o2}+R_s}$ $\approx \frac{-r_{o2}}{(1+\eta)R_s}$ $(\approx -R_{top}/R_{bot})$			
CD (buffer)	R_S		$A_v = \frac{g_{m1}}{g_{m1}+g_{mb1}+\frac{1}{R_s}+\frac{1}{r_{o1}}}$ $\approx \frac{g_{m1}}{g_{m1}+g_{mb1}}$ if $g_{m1} \gg \frac{1}{R_S} + \frac{1}{r_{o1}}$ $= \frac{1}{1+\eta}$ ≈ 1	$r_{s1} \| R_S$ $= \frac{1}{g_{m1}+g_{mb1}+\frac{1}{r_{o1}}} \| R_S$ $= \frac{1}{g_{m1}+g_{mb1}+\frac{1}{R_s}+\frac{1}{r_{o1}}}$	∞	
	CSL		$A_v = \frac{g_{m1}}{g_{m1}+g_{mb1}+\frac{1}{r_{o2}}+\frac{1}{r_{o1}}}$ $\approx \frac{g_{m1}}{g_{m1}+g_{mb1}}$ if $g_{m1} \gg \frac{1}{r_{o2}} + \frac{1}{r_{o1}}$ $= \frac{1}{1+\eta}$ ≈ 1	$r_{s1} \| r_{02}$ $= \frac{1}{g_{m1}+g_{mb1}+\frac{1}{r_{o2}}+\frac{1}{r_{o1}}}$	∞	CSL to avoid tradeoff between gain and voltage swing
CG	R_D		$A_v = \frac{r_{s1}}{R_S+r_{s1}}(\frac{1}{r_{o1}} + g_{m1} + g_{mb1})(r_{o1}' \| R_D)$ $\approx \frac{r_{s1}}{R_S+r_{s1}}(g_{m1} + g_{mb1})(r_{o1}' \| R_D)$ $r_{s1} = \frac{r_{o1}+R_D}{1+(g_{m1}+g_{mb1})r_{o1}} \approx \frac{1}{g_{m1}+g_{mb1}}$ $r_{o1}' = [1 + \frac{R_S}{r_{o1}} + (g_{m1} + g_{mb1})R_S]r_{o1}$ $r_{o1}' \approx [1 + (g_{m1} + g_{mb1})R_S]r_{o1}$	$r_{o1}' \| R_D$	r_{s1}	small input impedance, large output impedance, good for current amplifier

	CSL		$A_v = \frac{r_{s1}}{R_S+r_{s1}}(\frac{1}{r_{o1}} + g_{m1} + g_{mb1})(r_{o1}'\parallel r_{o2})$ $\approx \frac{r_{s1}}{R_S+r_{s1}}(g_{m1} + g_{mb1})(r_{o1}'\parallel r_{o2})$ $r_{s1} = \frac{r_{o1}+r_{o2}}{1+(g_{m1}+g_{mb1})r_{o1}} \approx \frac{2}{g_{m1}+g_{mb1}}$ $r_{o1}' = [1 + \frac{R_S}{r_{o1}} + (g_{m1} + g_{mb1})R_S]r_{o1}$ $r_{o1}' \approx [1 + (g_{m1} + g_{mb1})R_S]r_{o1}$	$r_{o1}'\parallel r_{o2}$	r_{s1}	CSL to avoid tradeoff between gain and voltage swing
Cascode Current reuse, reduce power draw Higher BW since input to output C is less	CSL		$A_V = -G_m(r_o'\parallel r_{o3})$ $A_V \approx -g_{m1}r_{o3}$ $G_m = \frac{(g_{m2}+g_{mb2}+\frac{1}{r_{o2}})r_{o1}g_{m1}}{1+(g_{m2}+g_{mb2}+\frac{1}{r_{o2}})r_{o1}} \approx g_{m1}$ no current gain from CG $r_o' = [1 + \frac{r_{o1}}{r_{o2}} + (g_{m2} + g_{mb2})r_{o1}]r_{o2}$ $r_o' \approx [1 + (g_{m2} + g_{mb2})r_{o1}]r_{o2}$ RMB: cascode boost the r_{o1} with g_{m2}, then r_{o2}	$r_o'\parallel r_{o3}$ $\approx r_{o3}$	∞	Vout > $V_{OD1} + V_{OD2}$ Current due to current source g_{m1} in M1 will split to r_{o1} and M2, majority of the current will pass through M2 appearing at the output ⇒ Gm ≈ g_{m1} Looking up the source of M2, $r_{s2} = \frac{r_{o2}+r_{o3}}{1+(g_{m2}+g_{mb2})r_{o2}} \approx \frac{2}{g_{m2}+g_{mb2}}$ This is small, so the voltage gain from input to X is small (compared to normal CS), this means the CGD1 is not Millered and amplified to be very big at the input. Another advantage of cascode, presenting lower input capacitance.

	Cascode CSL		$A_V = -G_m(r_o' \| r_o'')$ $A_V \approx \frac{-g_{m1}g_{m3}r_{o3}r_{o4}}{2} \quad for\ (r_o' = r_o'')$ $G_m = \frac{(g_{m2}+g_{mb2}+\frac{1}{r_{o2}})r_{o1}g_{m1}}{1+(g_{m2}+g_{mb2}+\frac{1}{r_{o2}})r_{o1}} \approx g_{m1}$ $r_o' = [1 + \frac{r_{o1}}{r_{o2}} + (g_{m2} + g_{mb2})r_{o1}]r_{o2}$ $r_o' \approx [1 + (g_{m2} + g_{mb2})r_{o1}]r_{o2}$ $r_o'' = [1 + \frac{r_{o4}}{r_{o3}} + g_{m3}r_{o4}]r_{o3}$ $r_o'' \approx [1 + g_{m3}r_{o4}]r_{o3}$	$r_o' \| r_o''$	∞	top headroom $V_{OD3} + V_{OD4}$ bottom headroom $V_{OD1} + V_{OD2}$ Current due to current source g_{m1} in M1 will split to r_{o1} and M2, majority of the current will pass through M2 appearing at the output $\Rightarrow Gm \approx g_{m1}$

- Always follow 2-Port Transconductance Amplifier Model when deriving
- If capacitors are involved, all above can be reused:
  - When finding Rin and Rout, which is now Zin Zout, bring the capacitance into R_S and R_D to form Z_S and Z_D
  - **But in the finding of transfer function the R and C could change and forms a LPF divider, NOT in parallel**